Lean CLEAN: Deconvolution Algorithm for Radar Imaging of Contiguous Targets

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High-resolution microwave images of contiguous targets often suffer from sidelobe artifacts. Deconvolution algorithms, such as Coherent CLEAN, do not perform well in these scenarios because of spurious peaks and peak mislocations. These spurious and mislocated peaks are often interpreted as real targets by the CLEAN algorithm. In this paper, we propose the Lean CLEAN technique that addresses this problem. The Lean CLEAN technique consists of a postprocessing block that works with the CLEAN algorithm. It detects and discards the spurious peaks and then consolidates the multiple small peaks corresponding to mislocated targets. Extensive simulations suggest that the proposed technique works well with different CLEAN techniques.

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I. INTRODUCTION

Reconstructed radar target images usually suffer from sidelobe artifacts in either the range or the angle domain. The net effect is that the dynamic range of the image is reduced. The sidelobes can be significantly reduced by a technique called CLEAN [1–3]. CLEAN is a deconvolution technique developed initially for radio astronomy. Since its introduction, several variants of the basic CLEAN algorithm can be found in the literature [4–7]. CLEAN has also been used as a denoising tool [8] and an optimum detector [9]. A modification to the basic CLEAN algorithm, called Coherent CLEAN, was proposed in [10] for radar target images. The Coherent CLEAN algorithm fails for images of contiguous targets with high sidelobes.

High range resolution radars require the transmitted signal to have a large bandwidth. However, jammers and legitimate users of the spectrum often pose problems with using a large bandwidth. One possible way to tackle these problems is to break up the wideband spectrum into a series of frequency subbands. The wideband spectrum can then be "thinned" by eliminating some sections of the spectrum [11]. Thinning the spectrum also results in less data acquisition and processing time, which is critical for real-time imaging. However, radar imaging with a thinned spectrum results in high sidelobes in the range domain. Sequence CLEAN was proposed to handle imaging with high sidelobes of closely spaced targets [12, 13]. Sequence CLEAN involves a tree search algorithm to find the best sequence in which to cancel out the peaks. Better performance is achieved at the cost of increased computational complexity.

The basic problem of radar imaging of contiguous targets is that the sidelobes interact, causing peak mislocations and spurious peaks. In this paper, we propose a postprocessing technique that tackles this problem. The rest of the paper is organized as follows: Section II describes the CLEAN algorithms used for radar imaging. In Section III, we discuss the problems of imaging contiguous targets. In Section IV, we introduce the concept of Lean CLEAN. In addition, the location estimator and the height estimator are analyzed in this section. The algorithm for Lean CLEAN is given in Section V. Section VI gives the simulation results for closely spaced targets. Finally, the paper concludes in Section VII.

II. CLEAN ALGORITHMS FOR RADAR IMAGING

In this section, we provide a brief background description of the various CLEAN algorithms for radar imaging. The CLEAN algorithm was originally developed by astronomers to increase the resolution of photographs of stars [1]. It works by identifying the brightest star in the image. The point spreading function (PSF) of the brightest star is then subtracted

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from the location of the star in the image. The PSF is the response to a point source. The location and intensity information for the brightest star are retained. Once the brightest star is removed, the procedure is repeated with the next brightest source in the image until the remaining scene is at the noise floor or some other predefined stopping criterion is applicable. The amplitude and position of each source are noted and represent the "cleaned" image.

In some cases, the point sources are convolved with the PSF minus the sidelobes to produce the cleaned image. The deconvolution technique was modified to work for radar images, which are essentially complex [10]. This modified CLEAN algorithm, called Coherent CLEAN, works by subtracting the complex PSF (radiation pattern or image response) weighted by the complex amplitude from the location of the brightest point. The new image is devoid of the strongest target, as well as its high sidelobes, thus exposing weaker targets that were previously submerged in the sidelobes.

Let the one-dimensional "dirty" image be represented by $s_0(x)$, where x represents the range dimension. Let the PSF (angle or range response) be denoted by $p(x)$. The first step of the Coherent CLEAN process removes a fraction γ of the energy of the brightest target source by forming

$$s_1(x) = s_0(x) - \gamma p(x - x_k) \tag{1}$$

where x_k is the location of the brightest point and $s_1(x)$ is the image with a part of the strongest target subtracted, along with its sidelobes. The loop gain γ is the fraction of the source amplitude subtracted during each iteration. The loop gain is chosen smaller than unity for stability of the CLEAN procedure. The Coherent CLEAN algorithm is based on the assumption that the scene consists of isolated and independent targets. However, many real-life targets are contiguous. Even if we approximate the contiguous targets by very closely spaced point sources, they can hardly be assumed to radiate independently. The sidelobes and the mainlobes of these closely spaced point sources can interact constructively and destructively, causing spurious peaks and peak mislocations. The problems of contiguous targets were addressed by the technique called Sequence CLEAN [12], which works with not just the brightest target but also the top m brightest targets. For $m = 1$, Sequence CLEAN is the Coherent CLEAN algorithm.

III. PROBLEMS OF IMAGING CONTIGUOUS TARGETS

Consider the problem of high sidelobes in range. Pulse compression is often used for high range resolution. High range sidelobes occur when, out of practical considerations, only a fraction of the frequency spectrum is used for image reconstruction [11, 14, 15]. A necessary condition for Coherent CLEAN, derived on lines similar to those given in [1], to ensure that the highest peak is the global maximum of the image is $M > B_{\mathrm{P}} Q e^{\sigma_\phi^2}(2 - e^{\sigma_\phi^2})$, where M is the number of frequency subbands used for pulse compression, Q is the number of closely spaced targets in the scene, B_{P} is the peak sidelobe estimator [16], and σ_ϕ^2 is the upper bound of the scene-dependent phase errors. For most problems, σ_ϕ^2 is a few tenths of a square radian [10]. B_{P} for a highly thinned spectrum is close to 2.5. Thus, the necessary condition becomes $M > 2.5Q$ [12]. From this analysis, the highest peak clearly is often a spurious peak for closely spaced or contiguous targets. The CLEAN algorithm, which tries to cancel out the highest peak first, often starts with a spurious peak, thereby introducing more errors in the process. The formation of spurious peaks is an outcome of the coherent interaction of the complex sidelobes of the closely spaced targets. These sidelobes themselves depend on the subbands of the thinned spectrum used for imaging.

The other problem is that of mislocated peaks. Because of the sidelobes of closely spaced targets and noise, the peak of the mainlobe is often displaced. The CLEAN algorithm works with the peak location of the highest peak. If the highest peak in the image is a true but displaced peak, the CLEAN algorithm runs into problems. This is because it subtracts out a portion of the PSF from an incorrect location. Because we are dealing with microwave images that consist of both amplitude and phase, the subtraction of the PSF from an incorrect location may result in the addition of target mass, thereby defeating the objective of the CLEAN technique. As a consequence, a true peak often shows up as a series of smaller peaks near the true target location when we process the input image of a contiguous target.

IV. LEAN CLEAN CONCEPT

As discussed in the previous section, images of contiguous targets are plagued with mislocated peaks and spurious peaks (as opposed to true peaks). A mislocated peak occurs when the actual location of a true target peak gets shifted because of interaction with the sidelobes of other closely spaced targets. Spurious peaks occur because of the coherent interaction of the sidelobes of closely spaced targets. The CLEAN algorithm often detects peaks that are either mislocated or spurious and then works with the erroneous location. Even the variants of the CLEAN technique, such as Sequence CLEAN, which work with the highest m peaks, rely on the peaks obtained from the image. If these peaks are spurious or mislocated, the problem persists. Because the CLEAN algorithm subtracts out in parts (dictated

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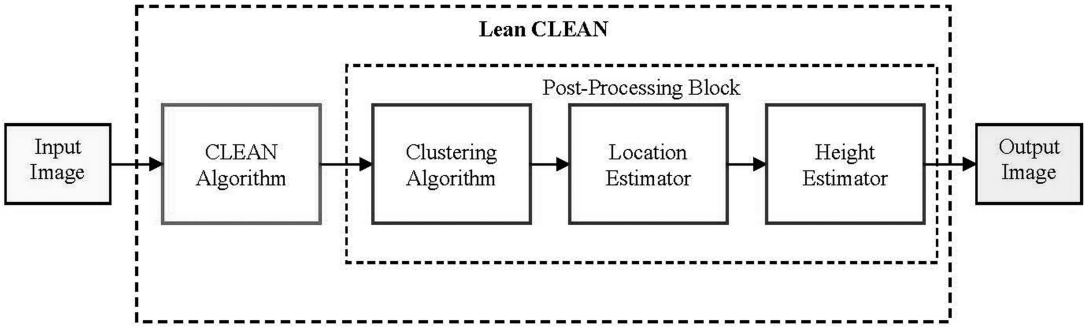

Fig. 1. Block diagram for Lean CLEAN technique.

by the loop gain γ), each subtraction step from an incorrect location may result in a different mislocated peak. Consequently, the reconstructed image consists of several small peaks near the true peak location. For a spurious peak, the CLEAN algorithm is unable to assess whether it is a true peak or not. In the reconstructed image, spurious peaks often show up as detected peaks.

The block diagram of the proposed Lean CLEAN technique is given in Fig. 1. The Lean CLEAN technique consists of a postprocessing block that works with the CLEAN algorithm. For simplicity, we describe it here in one dimension. However, the algorithm can be easily generalized to two dimensions. The postprocessing block of the Lean CLEAN technique performs the following steps on the reconstructed image produced by the CLEAN algorithm.

A. Clustering

The first step of the postprocessing block of the Lean CLEAN technique is to form clusters of the peaks detected by the CLEAN algorithm. We define a maximum separation distance $d_{\max}$. If the distance between any two detected peaks is less than $d_{\max}$, they are classified in the same cluster. As a result of the clustering process, the image now consists of clusters of peaks. The basic idea is that the peaks within the same cluster correspond to the same true peak. Because we are dealing with closely spaced targets, one possible choice can be $d_{\max} = 0.5\Delta R$, where ΔR represents the range resolution. Here, the closely spaced targets are separated by approximately ΔR, and the peaks separated by greater than $0.5\Delta R$ may belong to the adjacent target. Choosing $d_{\max} < 0.5\Delta R$ leads to fewer errors in peaks being clustered incorrectly but might leave more peaks unclustered. The choice of $d_{\max} < 0.5\Delta R$ also ensures that there is only one true peak in the same cluster.

B. Detection and Removal of Spurious Peaks

As discussed earlier, the multistep process of removal of a mislocated peak results in several small peaks located close to one another. The clustering subblock tries to collect all such peaks in the same cluster. However, this is not true for a spurious peak. The cluster corresponding to a spurious peak has few peaks. Thus, the number of peaks in a cluster can be used to classify whether a particular cluster corresponds to a true peak or a spurious peak. We define a minimum number $N_{\min}$. If the number of peaks in a cluster is less than $N_{\min}$, we declare it as a cluster corresponding to a spurious peak and eliminate it. Let the probability of occurrence of a spurious peak be P_{SP} and the minimum bin size along the range dimension be Δr. This bin size depends on the sampling rate of the imaging equipment. Here, $\Delta r < \Delta R$, and the maximum number of peaks possible within a cluster is $\lfloor d_{\max}/\Delta r \rfloor$. Assuming that the spurious peaks can occur at any of the bins with equal probability, the expected number of spurious peaks within a cluster defined by $d_{\max}$ can be written as

$$\overline{N_{\mathrm{SP}}} = \left\lfloor \left(\frac{d_{\max}}{\Delta r} \right) P_{\mathrm{SP}} \right\rfloor \leq \left\lfloor \left(\frac{d_{\max}}{\Delta r} \right) \right\rfloor \tag{2}$$

where the probability of the occurrence of spurious peak is given by [12]:

$$P_{\mathrm{SP}} = \int_1^{\infty} \frac{2r}{\sigma^2} e^{-r^2/\sigma^2} dr. \tag{3}$$

Here, $\sigma^2 = (1/M + 1/\mathrm{SNR})$, M is the number of frequency subbands used, and SNR corresponds to the signal-to-noise ratio. Here, $N_{\min} = \overline{N_{\mathrm{SP}}}$. As expected, the choice of $N_{\min}$ depends on the P_{SP}. As examples, for a synthesized bandwidth of 200 MHz, $\Delta R = 0.75$ m, and for a sampling rate of 3×10^9 per second, $\Delta r = 0.10$ m. If $d_{\max} = \Delta R = 0.75$ m and $P_{\mathrm{SP}} = 0.2$, then $\overline{N_{\mathrm{SP}}} = 1$ and a possible choice for $N_{\min}$ is 3.

C. Location Estimation of True Peaks

After the removal of the spurious peaks, we are left with clusters of peaks corresponding to true peaks. The ith cluster contains $N_i > N_{\min}$ peaks. We perform the location estimation of the true peak based on the locations of the peaks within the cluster. Let the location of the jth peak within the ith cluster be l_{ij}. Let $\hat{l}_i$ be the estimate of the location for the ith

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cluster. The mean square location error (MSLE) can be written as

$$\text{MSLE} = \frac{1}{N_i}\sum_{j=1}^{N_i}(\hat{l}_i - l_{ij})^2. \tag{4}$$

Minimizing the MSLE with respect to $\hat{l}_i$, we obtain

$$\hat{l}_i|_{\text{opt}} = \frac{1}{N_i}\sum_{j=1}^{N_i} l_{ij}. \tag{5}$$

Here, we are considering contiguous objects approximated by closely spaced point targets. By "closely spaced" we mean that the spacing between the targets is on the order of the range resolution ΔR. The range resolution is given by [17] as

$$\Delta R \approx \frac{c}{2B} \tag{6}$$

where c is the speed of light and B is the bandwidth used for imaging.

First, we derive a probabilistic upper bound on the location error. Let additive white Gaussian noise shift the mainlobe of the peak laterally by Δx. The mean of the random variable Δx is zero, and the variance is given by [18] as

$$\sigma^2_{\Delta x} = \frac{(\Delta R)^2}{\text{SNR}} \tag{7}$$

where SNR corresponds to the signal-to-noise ratio after pulse compression. Intuitively, if the range resolution is high (i.e., ΔR is small), the width of the mainlobe of the PSF would be small, resulting in a smaller Δx. Suppose the power in the pulse-compressed mainlobe is P_{ML} and the noise power is σ^2_N; then,

$$\text{SNR} = \frac{P_{\text{ML}}}{\sigma^2_N}. \tag{8}$$

Let the sidelobe of the PSF be represented by $l(x)$. The in-phase and quadrature components of $l(x)$ at any location can be assumed to be a complex Gaussian random variable with the mean $\mu_{\text{SL}} = 1/\sqrt{2}M$ and variance $\sigma^2_{\text{SL}} = 1/2M$ [17], where M is the number of frequency subbands used for imaging. Assume that the scene consists of Q closely spaced targets. Because the sidelobes of the Q interacting targets can be taken as independent, the sum of the Q sidelobes is a complex Gaussian random variable with the mean $Q\mu_{\text{SL}}$ and the variance $Q\sigma^2_{\text{SL}}$. Assuming the average sidelobe level is small in comparison to the mainlobe, the mislocated peak may be considered as resulting from an additive Gaussian noise with the variance

$$\sigma^2_{\text{eff}} = \sigma^2_N + Q\sigma^2_{\text{SL}}. \tag{9}$$

From (7) through (9), the variance of the random variable $\Delta x'$, which represents the lateral shift of the mainlobe peak because of the combined effect of the sidelobes and noise, can be written as

$$\sigma^2_{\Delta x'} = \frac{(\Delta R)^2(\sigma^2_N + Q\sigma^2_{\text{SL}})}{P_{\text{ML}}}. \tag{10}$$

From Chebyshev's inequality [19], we have the upper bound

$$P(|l_{\text{TP}} - \hat{l}_i| \geq \Delta R) \leq \left(\frac{\sigma^2_N + Q\sigma^2_{\text{SL}}}{P_{\text{ML}}}\right) \tag{11}$$

where l_{TP} is the true peak location and $\hat{l}_i$ is the estimate of the location for the ith cluster given by (5).

Next, we find the optimum location estimates. Let the height associated with the jth recovered peak within the ith cluster be h_{ij} and the average height of the ith cluster be $\bar{h}_i$. If we give more weight to the taller peaks, one way of expressing the objective function can be

$$\text{MSLE} = \frac{1}{N_i}\sum_{j=1}^{N_i}\left(\frac{h_{ij}}{\bar{h}_i}\right)(\hat{l}_i - l_{ij})^2. \tag{12}$$

Minimizing the weighted MSLE, as expressed in (12), with respect to $\hat{l}_i$, we obtain

$$\hat{l}_i|_{\text{opt}} = \left(\frac{1}{\sum_{k=1}^{N_i} h_{ik}}\right)\sum_{j=1}^{N_i} h_{ij} l_{ij}. \tag{13}$$

D. Height Estimation of True Peaks

This is the last stage of the postprocessing block, where the estimation of the height of the true peak is done. First, consider the following estimate of the height for the ith cluster:

$$\hat{\alpha}_i = \sum_{j=1}^{N_i} h_{ij} \tag{14}$$

where N_i represents the total number of peaks within the cluster and h_{ij} is the jth recovered peak within the ith cluster. Let the actual height of the true peak be h_{TP}. Let the total contribution of the sidelobes of other closely spaced targets that result in the formation of the jth peak in the ith cluster be s_{ij}. Let the contribution of noise be n. The CLEAN algorithm subtracts only a fraction of the peak, as dictated by the loop gain γ. Therefore, each small peak recovered within a cluster represents only a portion of the true peak. Thus, the height of the jth peak recovered in the ith cluster can be written as

$$h_{ij} = \gamma(h_{\text{TP}} + s_{ij} + n). \tag{15}$$

If we represent $z = s_{ij} + n$, then z is a Gaussian random variable with variance given by (9). From (14) and (15), we have

$$\hat{\alpha}_i = \gamma N_i h_{\text{TP}} + \gamma N_i z. \tag{16}$$

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If the algorithm is able to cancel out the entire true peak, then $\gamma N_i \approx 1$; i.e., the mislocated peaks within a cluster add up to the height of the true peak. Thus,

$$\hat{\alpha}_i = h_{\mathrm{TP}} + z. \tag{17}$$

Therefore, on average, $\hat{\alpha}_i$ overestimates the true height of the target in the ith cluster by $E[z] = Q\mu_{\mathrm{SL}}$. So, we modify (14) and use the following height estimator:

$$\hat{h}_i = \sum_{j=1}^{N_i} h_{ij} - Q\mu_{\mathrm{SL}}. \tag{18}$$

We already know Q from the stage where we eliminated the spurious peaks and retained only the true peaks. The average sidelobe level μ_{SL} can be estimated from the PSF obtained from an isolated target.

We call this the Lean CLEAN technique because it helps shed the excess baggage of the spurious peaks and consolidates the multiple small peaks corresponding to mislocated targets.

V. ALGORITHM FOR LEAN CLEAN

The technique for Lean CLEAN can be applied in one dimension or both dimensions simultaneously. The technique can be summarized as follows:

1) Choose a desired CLEAN algorithm, and perform the deconvolution to obtain a set of peaks.
2) Define a maximum separation distance $d_{\max}$, and perform clustering of peaks. If the distance between any two peaks is less than $d_{\max}$, they are classified in the same cluster.
3) Define a number $N_{\min}$. If the number of peaks in a cluster is less than $N_{\min}$, we declare it as a cluster corresponding to a spurious peak and eliminate it.
4) For a given cluster, estimate the location of the true peak by either (2) or (10).
5) For a given cluster, estimate the height of the true peak using (15).
6) Form the clean image by convolving the mainlobe of the PSF with properly sized δ functions at the estimated target locations.

In Step 1 of the algorithm, any CLEAN technique can be used. The end result depends on the quality of the output of the initial CLEAN technique. The Coherent CLEAN algorithm works with the highest peak, where as the Sequence CLEAN technique works with the highest m peaks at each stage. Coherent CLEAN is thus a special case ($m = 1$) of Sequence CLEAN, which involves an exhaustive tree search algorithm.

In Step 1 of the algorithm, it is also possible to have a compromise between the Coherent CLEAN algorithm and the Sequence CLEAN technique. Suppose we pick the highest m peaks at any stage of the algorithm. We cancel each of the m peaks by subtracting the PSF from these m locations one by one. We do not follow the mary tree for each peak canceled, as in Sequence CLEAN. Instead, in the next step, we again choose the highest m peaks and cancel them out. We continue the process until further subtraction leads to increase in the total target mass. We call this the Modified Coherent CLEAN technique. The algorithm for the Modified Coherent CLEAN technique is as follows:

1) At the kth step of the algorithm, pick the highest m peaks in the image $s_k(x)$ with the target mass $T_k = \int_{-\infty}^{\infty} s_k(x)s_k^*(x)dx$.
2) For the ith peak ($i = 1,2,\ldots,m$), subtract a fraction γ of the PSF $p(x)$ from the peak location, i.e., $s_k'(x) = s_k(x) - \gamma p(x - x_i)$.
3) Repeat Step 2 for each peak ($i = 1,2,\ldots,m$). The subtraction of peaks is carried out in no particular order. After subtracting out the m peaks, obtain the partially cleaned image $s_{k+1}(x)$.
4) Compute the target mass T_{k+1}. If $T_{k+1} < T_k$, go to Step 1; otherwise, develop a list of peak locations and the respective peak amplitudes of the targets canceled.
5) Form a cleaned image by convolving the mainlobe of the PSF with properly sized δ functions located at the recovered target locations.

As expected, the computational complexity of this Modified Coherent CLEAN technique is much less than that of Sequence CLEAN (grows linearly with m). As depicted in Fig. 1, the proposed Lean CLEAN technique works on the output of a particular CLEAN algorithm. It could be Coherent CLEAN, Sequence CLEAN, or Modified Coherent CLEAN.

VI. SIMULATION RESULTS

Exhaustive simulations were carried out to test the proposed algorithm. As mentioned earlier, the proposed algorithm works with the output of any CLEAN technique. It has been tested with Coherent CLEAN, Sequence CLEAN, and Modified Coherent CLEAN, which is the compromise between Coherent CLEAN and Sequence CLEAN. Here, we give one of the typical scenarios as a representative example with the Modified Coherent CLEAN technique. The simulated scene of Fig. 2 consists of six targets, of which five are closely spaced in range. The targets are of unequal heights (1.0, 1.0, 1.1, 0.9, 1.0, and 0.9). Pulse compression has been used to synthesize a bandwidth of 200 MHz, giving a range resolution of $\Delta R = 0.75$ m. The first target is isolated from the cluster of the other five targets. The average spacing between the five closely located targets is 1.5 m. For a bandwidth of 200 MHz, these five targets are just resolvable. Thus, these point sources closely simulate a contiguous target. We use a thinned spectrum to image these closely spaced targets. A

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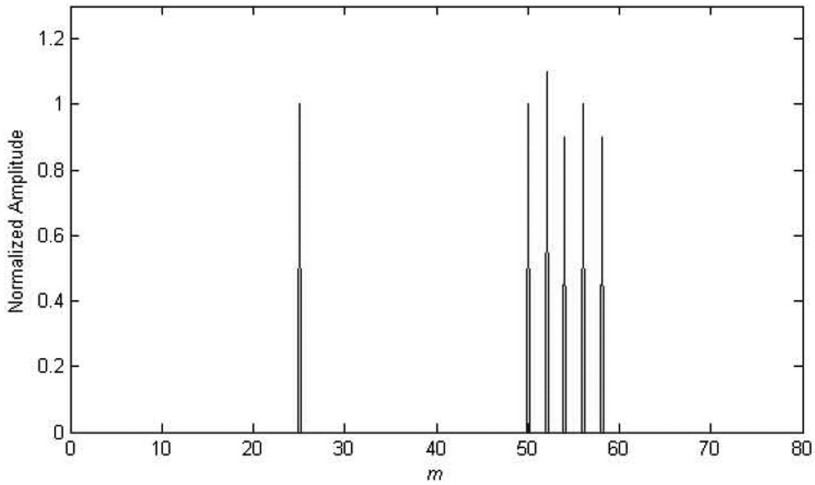

Fig. 2. Location of targets in simulated scene.

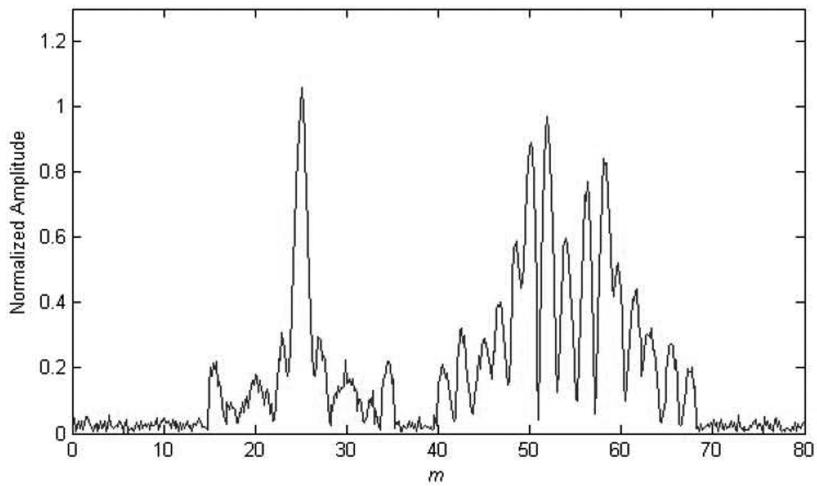

Fig. 3. Reconstructed images (range profile) corresponding to target scene shown in Fig. 2. Here, spectral thinning $T = 30\%$ and $Q = 5$.

thinned spectrum also results in a PSF with high sidelobes [12]. Specifically, we have used 30% spectral thinning. The reconstructed one-dimensional image (range profile) is given in Fig. 3. The PSF used in the experiment was obtained from the image of the isolated target. It can be seen that the waveform used results in very high sidelobes. Noise of −20 dB has been added to the raw data to simulate realistic conditions.

Modified Coherent CLEAN, described in Section V, was then applied to the reconstructed image (range profile), as depicted in Fig. 3. In this case, $m = 4$; i.e., the algorithm worked with the highest four peaks at any stage. For $m = 1$, Modified Coherent CLEAN becomes the regular Coherent CLEAN algorithm. The loop gain $\gamma = 0.3$. Fig. 4 shows the target peaks recovered by the Modified Coherent CLEAN technique. It can be seen that Modified Coherent CLEAN was not able to recover the target locations and heights correctly. The technique also generated several spurious peaks and mislocated peaks.

The peak locations and heights as recovered by Modified Coherent CLEAN were then processed by the clustering subblock of the Lean CLEAN technique. The parameters chosen were $d_{\max} = 0.3$ m ($0.4\Delta R$) and $N_{\min} = 3$. The clustering subblock discarded all clusters that contained fewer than three peaks. Fig. 5 shows the output of the Lean CLEAN technique. The Lean CLEAN technique was able to eliminate all spurious peaks and consolidate the mislocated peaks. To estimate the true peak locations, the weighted minimum mean square error (MMSE) estimator given in (13) was used. The average location error in this case was 0.12 m ($0.16\Delta R$). For the MMSE estimator given by (5), the average location error comes out to be 0.15 m ($0.21\Delta R$). Simulations were carried out for more than 1000 cases with different number of targets in the scene and different sidelobe levels of the PSF. In general, the weighted

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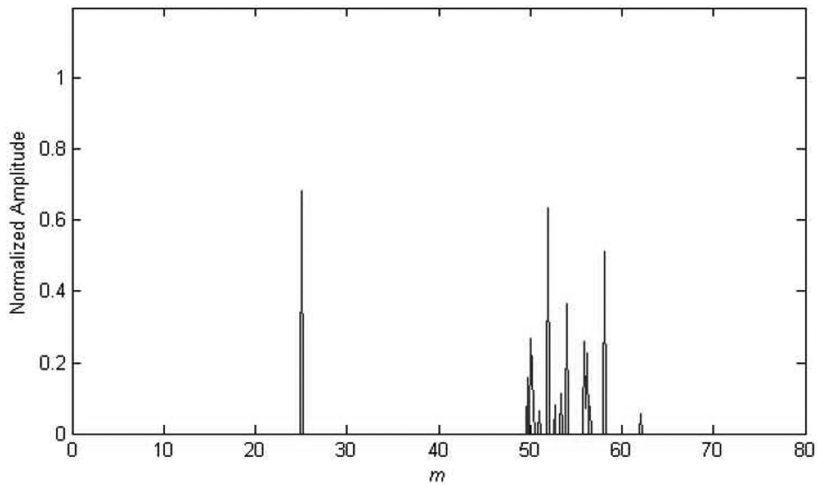

Fig. 4. Target locations and heights as recovered by Modified Coherent CLEAN.

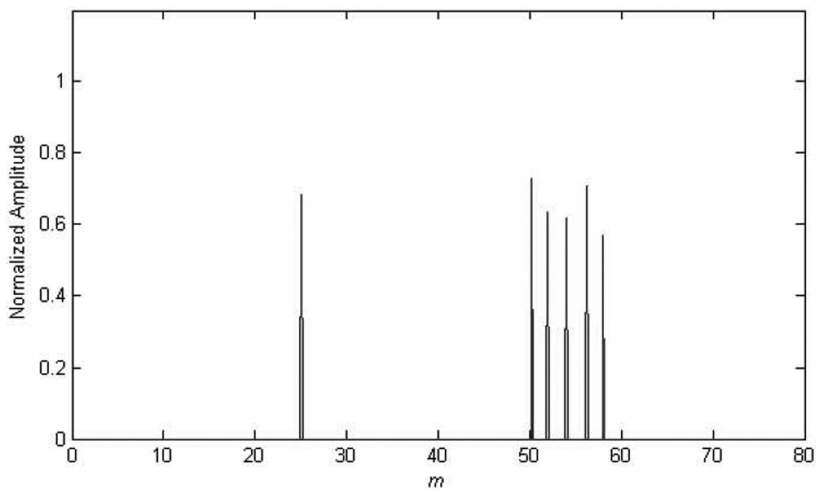

Fig. 5. Target locations and heights as recovered by Lean CLEAN.

MMSE estimator described by (13) works better than the MMSE location estimator described by (5).

The height estimator given by (18) was used to determine the height of the targets. The isolated peak, located 25 m downrange, was handled as a special case in a preprocessing stage. In this preprocessing stage, the PSF was extracted from the isolated target and used for the Modified Coherent CLEAN technique. The accuracy of the height estimation of the targets depends on the noise present in the image, as well as the choice of the loop gain factor γ. The height estimates can be affected if the algorithm stops before the entire target mass is subtracted out. In the simulations, the average error in the height estimation was found to be within a range of 10%–20%.

The proposed technique has also been tested on two-dimensional (2-D) images with encouraging results. We present here one typical result of Lean CLEAN in 2-D. Fig. 6(a) shows the overall scene being imaged. It consists of one isolated target and 25 closely spaced targets, representing a contiguous target. The spacing between these closely spaced targets is approximately 1 m, as depicted in the zoomed-in image given in Fig. 6(b). These targets are of varying heights. Noise of −20 dB has been added to the raw data to simulate realistic conditions. The PSF used in the experiment was obtained from the 2-D image of the isolated target. The reconstructed 2-D image using Modified Coherent CLEAN is given in Fig. 6(c). As observed from the figure, the locations and the heights have not been properly estimated, and there are several spurious peaks. The Lean CLEAN technique was able to eliminate all spurious peaks and consolidate the mislocated peaks. To estimate the true peak locations, the weighted MMSE estimator given in (13) was used. The average location error in this case was 0.18 m ($0.24\Delta R$). The output of the Lean CLEAN block is shown in Fig. 6(d).

The performance of Lean CLEAN in the presence of different noise levels has also been studied. The plot of the average MSLE (in units of ΔR) versus that of the SNR (in decibels) is given in Fig. 7. The scene consists of five closely spaced targets and one isolated target, as shown in Fig. 2. The range profile has been

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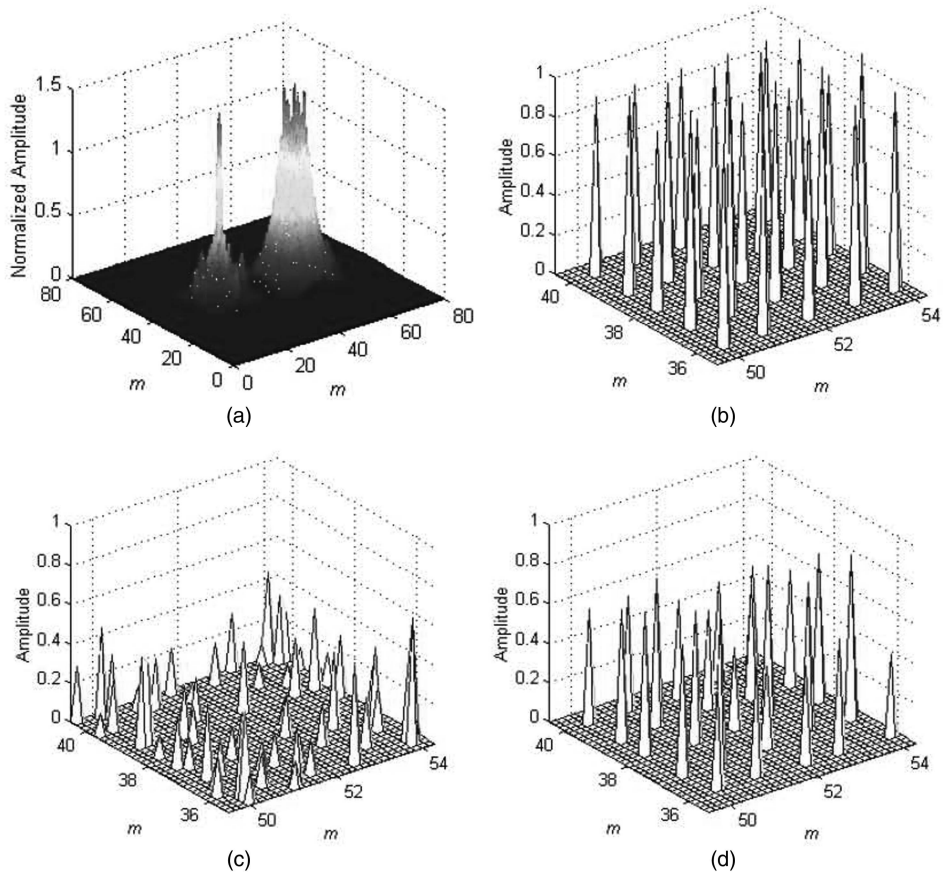

Fig. 6. (a) 2-D scene. (b) Locations and heights of the closely spaced targets in 2-D. (c) Target locations and heights as recovered by Modified Coherent CLEAN. (d) Target locations and heights as recovered by Lean CLEAN.

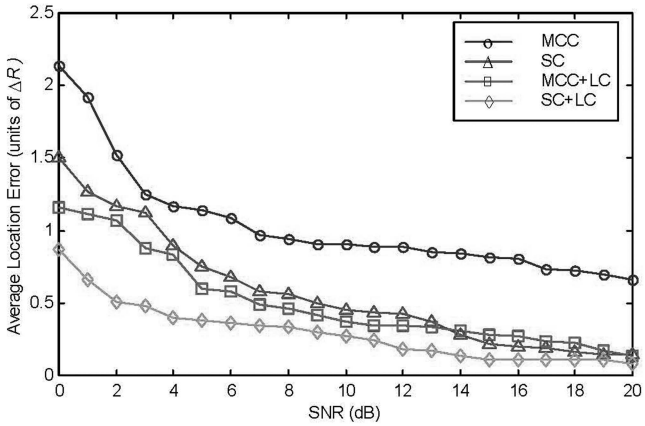

Fig. 7. Noise performance of Lean CLEAN for five closely spaced targets. Plots are given for Modified Coherent CLEAN (MCC), Sequence CLEAN (SC), Lean CLEAN applied to output of MCC (MCC+LC), and Lean CLEAN applied to output of SC (SC+LC).

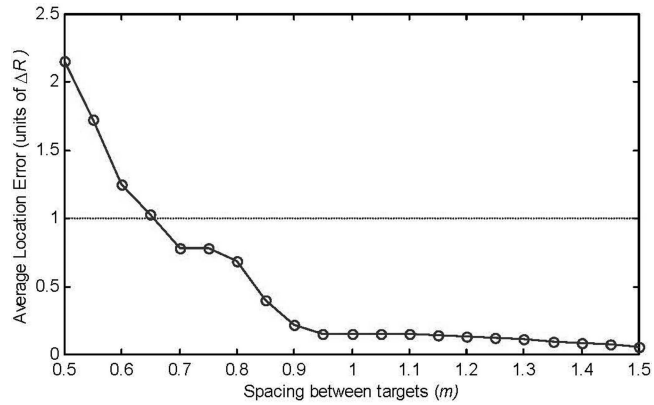

Fig. 8. Effect of spacing between closely spaced targets.

constructed with spectral thinning of 30%. Plots are given for Modified Coherent CLEAN, Sequence CLEAN, Lean CLEAN applied to the output of Modified Coherent CLEAN, and Lean CLEAN applied to the output of Sequence CLEAN. For Modified Coherent CLEAN and Sequence CLEAN, the number of highest peaks chosen at each step of the algorithm is $m = 5$. Each point in the figure has been averaged 100 times. From Fig. 7, we observe that Lean CLEAN improves the performance in low SNR scenarios. Intuitively, the performance of the Modified Coherent CLEAN and the Sequence CLEAN techniques degrades in low SNR because many peak locations are incorrect to begin with.

The effect of spacing between closely spaced targets on the average MSLE is shown in Fig. 8. The scene consists of five closely spaced targets. The spacing between the targets varies from $0.67\Delta R$ to

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$2\Delta R$, where range resolution ΔR is 0.75 m. Each point in the figure has been averaged 100 times. The figure shows that the proposed technique works almost up to the range resolution.

VII. CONCLUSIONS

In this paper, we have proposed a novel technique that addresses the problem of mislocated peaks and spurious peaks when the CLEAN algorithm is used for images of contiguous targets. Our technique can work with any existing version of the CLEAN algorithm. The proposed technique first forms clusters of the peaks and then eliminates the probable spurious peaks. It then estimates the true peak locations and heights. In essence, the Lean CLEAN technique discards the spurious peaks and consolidates the multiple small peaks corresponding to mislocated targets, thereby yielding a cleaned image. Extensive simulations have been carried out to test the proposed technique. For 30% spectral thinning with five closely spaced targets, the average location error was found to be $0.16\Delta R$, where ΔR is the range resolution, and it was possible to eliminate all spurious peaks. The proposed technique has also been tested on 2-D images with encouraging results. The Lean CLEAN technique may be used to further improve the performance of 2-D CLEAN techniques as applied to inverse synthetic aperture radar imaging [20].

REFERENCES

[1] Hogbom, J. A.
Aperture synthesis with non-regular distribution of interferometer baselines.
Astronomy & Astrophysics Supplement Series, **15** (1974), 417–426.

[2] Clark, B. G.
An efficient implementation of the algorithm CLEAN.
Astronomy & Astrophysics, **89** (1980), 377–378.

[3] Schwarz, U. J.
Mathematical-statistical description of the iterative beam removing technique (method CLEAN).
Astronomy & Astrophysics, **65** (1978), 345–356.

[4] Steer, D. G., Dewdney, P. E., and Ito, M. R.
Enhancements to the deconvolution algorithm "CLEAN."
Astronomy & Astrophysics, **137** (1984), 159–165.

[5] Segalovitz, A. and Frieden, B. R.
A "CLEAN"-type deconvolution algorithm.
Astronomy & Astrophysics, **70** (1978), 335–343.

[6] Schwarz, U. J.
The method "CLEAN" use, misuse and variations.
In C. van Schooneveld (Ed.), *Image Formation from Coherence Functions in Astronomy*, Dordrecht, The Netherlands: D. Reidel, 1979, 261–275.

[7] Deng, H.
Effective CLEAN algorithms for performance-enhanced binary coding signal detection.
IEEE Transactions on Signal Processing, **52**, 1 (2004), 72–78.

[8] Solo, V.
A modified CLEAN algorithm does L1-denoising.
In *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing*, Las Vegas, NV, Mar. 2008, 3665–3668.

[9] Foreman, T. L.
Reinterpreting the CLEAN algorithm as an optimum detector.
In *Proceedings of the International Conference on Radar*, Shanghai, China, 2006, 769–775.

[10] Tsao, J. and Steinberg, B. D.
Reduction of sidelobe and speckle artifacts in microwave imaging: The CLEAN technique.
IEEE Transactions on Antennas and Propagation, **36** (Apr. 1988), 543–556.

[11] Freedman, A., Bose, R., and Steinberg, B. D.
Thinned stepped frequency waveforms to furnish existing radars with imaging capability.
IEEE Aerospace and Electronic Systems Magazine, (Nov. 1996), 39–43.

[12] Bose, R., Freedman, A., and Steinberg, B. D.
Sequence CLEAN: A modified deconvolution technique for microwave images of contiguous targets.
IEEE Transactions on Aerospace and Electronic Systems, **38**, 1 (Jan. 2002), 89–97.

[13] Bose, R.
Sequence CLEAN technique using breeder genetic algorithm for contiguous radar target images with high sidelobes.
IEEE Transactions on Aerospace and Electronic Systems, **39**, 1 (Jan. 2003), 368–373.

[14] Bose, R., Freedman, A., and Steinberg, B. D.
High resolution 2-D imaging with spectrally thinned wideband waveforms.
In *Proceedings of the 2nd International Conference on Ultra-Wideband Short Pulse Electromagnetics*, Brooklyn, NY, Apr. 1994.

[15] Freedman, A., Bose, R., and Steinberg, B. D.
Techniques to improve the CLEAN deconvolution algorithm.
Journal of the Franklin Institute, **332B** (Sept. 1995), 535–554.

[16] Steinberg, B. D.
The peak sidelobe of the phased array having randomly located elements.
IEEE Transactions on Antennas and Propagation, **AP-20** (Mar. 1972).

[17] Steinberg, B. D.
Principles of Aperture and Array System Design.
New York: Wiley, 1976.

[18] Levanon, N.
Radar Principles.
New York: Wiley, 1988.

[19] Papoulis, A.
Probability, Random Variables, and Stochastic Processes, 3rd ed.
New York: McGraw-Hill, 1991.

[20] Berizzi, F., Martorella, M., and Acito, N.
Statistical CLEAN technique for ISAR imaging.
IEEE Transactions on Geoscience and Remote Sensing, **45**, 11 (Nov. 2007), 3552–3560.

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